Md Nafis Walid

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Career Objectives & Profiles

Embedded Software Engineer with hands-on experience in C-based firmware development for embedded networking devices. Strong foundation in embedded programming, networking protocols, system debugging, and problem-solving. Seeking to design, develop, and maintain reliable C/C++ embedded systems while continuously advancing my expertise in embedded software and computer architecture.

Educational Qualification

Bachelor of Science in Computer Science & Engineering February, 2019 – May, 2024
- Rajshahi University of Engineering & Technology (RUET) — Kazla, Rajshahi
- CGPA: 3.32/4.00

Higher Secondary Certificate (HSC) from Science Group July, 2016 – September, 2018
- Milestone College — Uttara, Dhaka-1230
- GPA: 5.00/5.00

Professional Experience

Research & Development Engineer
Shanghai BDCOM Information Technology Co., Ltd. (Bangladesh Office) May, 2024 – Present
- Developed and maintained C-based firmware for GPON/EPON ONU and Ethernet switch products on Linux- and VxWorks-based embedded platforms.
- Designed and implemented a TFTP-based firmware upgrade solution for embedded Ethernet switch devices.
- Implemented, customized, and validated networking features including DHCPv4, DHCPv6, PPPoE, IGMP, MLD, OMCI, TR-069, VLAN, and Ethernet switching.
- Customized firmware for international ISP customers and ensured interoperability with Huawei and ZTE OLTs.

Projects

4-bit CPU Design [GitHub]
- Designed and implemented a fully functional 4-bit CPU at the register-transfer level, including an ALU, register file, RAM, program counter, status flags, and an FSM-based control unit.
- Designed a custom instruction set architecture (ISA) and assembly language, implementing the complete instruction fetch–decode–execute cycle.

Network Packet Management [GitHub]
- Developed a packet management system in C to parse captured UDP packets into linked-list and hash-table data structures with reference counting and efficient packet lookup.
- Applied low-level memory management, pointer manipulation, and data structure optimization for packet processing.

Student Information Management [GitHub]
- Developed a console-based student information management system in C using doubly linked lists and modular programming.
- Implemented file-based persistent storage with complete CRUD operations for departments, students, and academic records.

Skills

Technical Skills

- **Programming Languages:** C, C++, Embedded C, Python, Assembly (8086), Java, Matlab
- **Embedded Systems & Firmware:** Embedded Firmware Development, Linux, VxWorks, GPON/EPON, Ethernet Switching, TFTP, CLI
- **Networking Protocols:** TCP/IP, UDP, IPv4/IPv6, DHCPv4, DHCPv6, PPPoE, IGMP/MLD Snooping, OMCI, TR-069, VLAN, Multicast
- **Web & Databases:** HTML, CSS, Bootstrap, JavaScript, MySQL, PostgreSQL
- **AI & Machine Learning:** Deep Neural Networks (DNN), Convolutional Neural Networks (CNN), Machine Learning, Computer Vision
- **Development Tools & Workspaces:** SVN, Git/GitHub, Wireshark, SecureCRT, GDB, GNU Make, Cisco Packet Tracer, Logisim, Digital, Tinkercad, Figma, Adobe XD, Adobe Illustrator, Adobe Photoshop, Microsoft Office
- **Core Concepts:** Computer Architecture, Operating Systems, Data Structures & Algorithms, Digital Logic Design, Microprocessors & Microcontrollers, Compiler Design, Hardware-Software Co-Design, Object-Oriented Programming (OOP)

Problem Solving / Competitive Programming

- **Codeforces:** Nafis_Walid – 380+ problems solved, 60+ contests
- **LeetCode:** Nafis_Walid – 100+ problems solved
- **LightOJ:** nafis_walid

Languages & Soft Skills

- **Languages:** Bengali (Native), English (Professional Working Proficiency)
- **Soft Skills:** Problem Solving, Critical Thinking, Leadership, Team Collaboration, Technical Communication, Event Management